Olive oil (virgin, refined, pomace): HMTc Standard

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CORRESPONDENCE

Corrections are logged publicly on the article page, each with its date, the affected value, and the source that resolved it.

KEYWORDS

heavy metals, food safety, dietary exposure

PUBLICATION HISTORY

This revision 2026-08.

Abstract

The Heavy Metal Tested & Certified standard for Olive oil (virgin, refined, pomace) (Culinary Oils and Fats), stated as sold. Each analyte carries the strictest applicable government maximum level in the product's native basis, with a disclosed read-across or a mandatory control where no government regulates the exact product.

1 | How every number on this standard is set

HMTc Method v2.0 (government-limit default; face value, no reconstitution; read-across / control / not-material for unregulated cells; no occurrence percentiles)

2 | Master limit table

All values in ppb (µg/kg), as sold basis. A brand meeting every value qualifies for the mark.

ANALYTE	HMTC LIMIT	BASIS
Lead (Pb)	80	government maximum level (GSO)
Cadmium (Cd)	30	government maximum level (Israel)
Total arsenic (tAs)	100	government maximum level (Codex)
Inorganic arsenic (iAs)	≤ tAs	bounded by tAs (reflex speciation)

Mercury, total (tHg)	20	government maximum level (Israel)
Methylmercury (MeHg)	≤ tHg	bounded by tHg (reflex speciation)
Nickel (Ni)	100	read-across (EU 2024/1987)
Tin (Sn)	200,000	read-across (EU 2023/915)
Aluminium (Al)	1,000	read-across (EU 10/2011)
Total chromium (Cr)	300	read-across (GB 2762)
Hexavalent chromium (Cr-VI)	10	read-across (CA SWRCB)

3 | Derivation — where each number comes from

Every cell traces to one named sovereign maximum level (the strictest for the nearest analog), or to an explicit control, not-material, or species-bounded determination. No occurrence statistic selects a certification value.

ANALYTE	HMTC	SOVEREIGN ANALOG / DETERMINATION
Lead (Pb)	80	government maximum level (GSO); Edible fats and oils
Cadmium (Cd)	30	government maximum level (Israel); Fats, oils, and fat emulsions (water-in-oil type)
Total arsenic (tAs)	100	government maximum level (Codex); Named vegetable oils
Inorganic arsenic (iAs)	≤ tAs	bounded by tAs (reflex speciation)
Mercury, total (tHg)	20	government maximum level (Israel); Fats, oils, and fat emulsions (water-in-oil type)
Methylmercury (MeHg)	≤ tHg	bounded by tHg (reflex speciation)
Nickel (Ni)	100	read-across (EU 2024/1987); infant/young-child formula, liquid
Tin (Sn)	200,000	read-across (EU 2023/915); canned foods other than canned beverages, except canned dried and canned powdered products
Aluminium (Al)	1,000	read-across (EU 10/2011); plastic food-contact materials and articles
Total chromium (Cr)	300	read-across (GB 2762); raw, pasteurized, sterilized, modified and fermented milk (GB 2762-2022 Table 7, section 4.7); total Cr, not Cr-VI

Hexavalent chromium (Cr-VI)	10	read-across (CA SWRCB); drinking water

4 | Cite this standard

Pendergrass, K. (2026). *Olive oil (virgin, refined, pomace): HMTc standard* (Standard No. 10.83138/hmtc.std.30.olive-oil). Institute of Contaminant Standards. https://doi.org/10.83138/hmtc.std.30.olive-oil

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References

[1] GSO
Binds Pb. Edible fats and oils.
MAXIMUM LEVEL

[2] Israel
Binds Cd, tHg. Fats, oils, and fat emulsions (water-in-oil type).
MAXIMUM LEVEL

[3] Codex
Binds tAs. Named vegetable oils.
MAXIMUM LEVEL

[4] EU 2024/1987
Binds Ni. infant/young-child formula, liquid.
READ-ACROSS

[5] GB 2762
Binds Cr. raw, pasteurized, sterilized, modified and fermented milk (GB 2762-2022 Table 7, section 4.7); total Cr, not Cr-VI.
READ-ACROSS

[6] CA SWRCB
Binds Cr-VI. drinking water.
READ-ACROSS

[7] EU 2023/915
Binds Sn. canned foods other than canned beverages, except canned dried and canned powdered products.
READ-ACROSS

[8] EU 10/2011
Binds Al. plastic food-contact materials and articles.
READ-ACROSS

STATUS AND LICENCE

A normative certification standard, editorially reviewed and version-controlled in-house; every value traces to a named sovereign maximum level, disclosed in the derivation. © 2026 Institute of Contaminant Standards / Heavy Metal Certified. All rights reserved.

COMPETING INTERESTS

This standard is published by the Heavy Metal Tested & Certified (HMTc) program, operated by the Institute of Contaminant Standards. Its statements about government regulations, toxicology, and occurrence are sourced to the cited primary and expert-body records and to the independently maintained Heavy Metal Index; the program's

own certification thresholds are not used as evidence for any factual claim made here.

DATA AVAILABILITY

The works this standard draws on are listed in full in the references above and on the standard page at https://heavymetalcertified.com/standards/olive-oil. Primary legal instruments and expert-body assessments are named with their public identifiers; the underlying occurrence and toxicology corpus is maintained on the Heavy Metal Index.

PUBLICATION HISTORY

This revision 2026-08. The full dated record is maintained at https://heavymetalcertified.com/standards/olive-oil.